

Griffin Laprise

Electrical Engineering Student
at Western University

226-626-1559
griffin.laprise@gmail.com
www.linkedin.com/in/griffinlaprise

EDUCATION

Bachelor of Engineering Science (BESc): Third Year Undergraduate, **Electrical Engineering**

2023 – 2027

Western University – London, ON

- Relevant courses: Digital Logic Systems, Electronic Circuits, Control Systems, Electric Machines, Power Systems 1, Signal Processing, Design of Electronic Systems

CO-CURRICULARS

Electrical Third Year Representative – Undergraduate Engineering Society at Western University

October 2025 – Present

- Acted on behalf of third-year Electrical Engineering students by sharing academic concerns, comments, and suggestions with the department and faculty.
- Functioned as a link between students and student councils with respect to communication about updates, events, and other opportunities.
- Advocated for the needs of the student body while supporting an initiative to improve the student experience in the Electrical Engineering department.

Vice President of Logistics – Canadian Engineering Leadership Conference 2025 Organizing Committee

April 2025 – January 2026

- Coordinated logistics for a multi-day engineering conference with over 240 attending students, and planned all venues, schedules, and resource allocation.
- Handled the distribution of funds from a \$120,000 conference budget to support operations and logistics.
- Worked with various teams to integrate timelines, technology requirements, and event dynamics.
- Conducted operations at the venue, fixing last-minute issues that might arise when hosting events.
- Supported a professional experience for attendees, speakers, and sponsors.

Electrical Lead – Western Engineering Electrium Mobility Design Team

September 2024 – April 2025

- Established a new undergraduate design team for Western Engineering with 30 active members that focused on creating cost-effective and ecologically friendly personal electric vehicles.
- Approval from the engineering undergraduate society and obtained \$2.5k in funding for the team.
- Oversaw and planned team meetings as well as workshops to guarantee the build and instructional seminars ran smoothly.
- Managed the Electrical, and Firmware sub teams, aiding in designs, providing advice, and making sure all sub teams worked effectively and adhered to a schedule.
- Used a variety of metal and wood machines for fabrication, SolidWorks for CAD, and Altium for PCB design.

WORK EXPERIENCE

Greenhouse General Labourer – Oasis Greenhouses Ltd

May July 2018 – Present

- Seeding and transplanting various fruits and vegetables.
- Oversaw the seeding line which was vital for not only providing each plant with enough soil but also a single seed per cell.
- Performed general maintenance for equipment around the facility such as tractors and electrical work on the seeding machine.

Field Inspector – Pride Seeds

June 2025 – September 2025

- Carried out thorough pollination counts and growth evaluations in several corn fields to assess the performance and development rates of various hybrid strains.
- Gathered and documented information on the growth of both male and female plants, detecting the presence of diseases and evaluating resistance to environmental stressors and drought.
- Autonomously oversaw field-to-field inspections, guaranteeing precise data gathering, organization, and reporting to assist breeding decisions and seed performance analysis.

Midnight Pea Combine Operator – Beneteau-Agri-Service

June 2024 – August 2024

- Operated an Oxbo pea harvester and performed general maintenance.
- Devoted 12-hour days 7 days a week to ensure all fields were harvested in time and prevent any of the crop from rotting.

SKILL SUMMARY

- Bilingual in both French and English.
- Experience in various leadership positions that require sufficient time management and teamwork skills.
- Experience in problem-solving through analyzing complex technical challenges, identifying root causes, and implementing effective, data-driven solutions.
- Experience in cross-functional collaboration with multidisciplinary teams to deliver projects efficiently and on schedule.
- Proficient with Microsoft Office Suite (ex. Word, Excel, PowerPoint).
- Valid G drivers license

Design Onshape, BOM creation, SolidWorks

Software Python, C++, Java, MATLAB

Electrical Arduino, Altium, Eagle, Quartus prime, Microcap, PCB design (THT and SMD), Soldering, Siemens TIA Portal

Mechanical FDM 3D printing, SolidWorks, Lathe, Mill, Rapid Prototyping, R&D, Material Selection

PROJECTS

Brushless DC Motor – Electric Machines Design Project

- Designed and fabricated a custom brushless DC outrunner motor within strict electrical and dimensional constraints, achieving stable operation at ~2500 RPM with smooth startup and low vibration.
- Performed electromagnetic and mechanical analyses to determine torque output, flux density, cogging effects, and winding parameters, resulting in an optimized 6-slot, 8-pole motor design.
- Collaborated in a team of four to develop detailed CAD models of the stator and rotor, optimizing air gap, bearing tolerances, and magnet placement for mechanical integrity and strong magnetic coupling.
- Assembled the motor, including three-phase coil winding in a delta configuration and precise installation of neodymium magnets.
- Tested and validated motor performance, confirming efficient operation, low acoustic noise, no thermal issues, and reliable long-term functionality.

AntiSpark PCB / Inrush Current Limiter – Western Engineering Electrium Mobility (WEEM)

- Personally designed an AntiSpark PCB for WEEM's electric skateboard, a student design team building cost-effective, eco-friendly personal electric vehicles.
- Engineered the circuit to limit inrush current when the battery is first connected, preventing the current spike that can arc and pit connector contacts.
- Used the transient voltage buildup across a pre-charge resistor to gradually reach the gate threshold voltage needed to turn on a pair of MOSFETs sequentially, letting them open one after another rather than switching instantly.
- Protected the ESC and other downstream components from spark-induced damage and stress, improving the reliability and lifespan of the vehicle's electrical system.
- Designed the full PCB in Altium Designer, from schematic capture through layout and routing.

Spot Welder Motor Controller – Electrical Design project

- Collaborated on a team project to automate a spot welder, designing a controller that managed the force, voltage, current, and position of the motor driving the electrode during welding.
- Built a custom GUI enabling users to set force and position limits and monitor live plots of all four input signals in real time.
- Implemented high-speed data logging, capturing 40kHz of data across the 0.2-second duration of each weld for post-weld analysis.
- Designed a custom PCB to route all sensor inputs into the microcontroller, including differential amplifiers for the load cells to accurately capture force readings.
- Incorporated clamping diodes to protect the circuit by keeping voltages within a safe operating range.
- Added anti-aliasing filters on all four input channels to preserve signal integrity and ensure clean, accurate logged data.

Personal Interests:

Repairing Watches, Woodworking, Agriculture, Pickle Ball, Golf, Astronomy